



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai 2026



ASSESSMENT TOOL - 3

CO2_AT3

Name : P Giriprasad Reddy

Register No : 192525172

Course code : CSA1613

Course name: Data Warehousing and Data Mining

Faculty name: Dr Kanimozhi K.V

SIMATS ENGINEERING

Assessment Tool 3

Course Code: CSA16

Course Name: Data Warehousing and Data Mining

Name : P Giriprasad Reddy

Reg No:192525172

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

Missing values identified: Age is missing for C2; Monthly_Spend is missing for C3; Purchase_Frequency is missing for C5.

(i) Median Imputation

Age (known values, sorted): 23, 27, 31, 36, 40, 42, 105 → median = 36. So Age(C2) = 36.

Monthly_Spend (known values, sorted): 20, 25, 35, 38, 45, 300, 500 → median = 38. So Monthly_Spend(C3) = 38.

Purchase_Frequency (known values, sorted): 5, 8, 9, 10, 12, 45, 50 → median = 10. So Purchase_Frequency(C5) = 10.

(ii) k-NN Imputation (k = 2, using Loyalty_Level / similar records as neighbors)

C2 (Age missing, Medium loyalty, freq = 8): nearest neighbours by similar Monthly_Spend & Purchase_Frequency are C1 (age 23) and C7 (age 36) → predicted Age $\approx (23 + 36) / 2 = 29.5 \approx 30$.

C3 (Monthly_Spend missing, High loyalty): nearest neighbours in the same loyalty class are C4 (500) and C8 (300) → predicted Spend $\approx (500 + 300) / 2 = 400$.

C5 (Purchase_Frequency missing, Low loyalty): nearest neighbours C1 (freq 5) and C7 (freq 9) → predicted Frequency $\approx (5 + 9) / 2 = 7$.

(iii) Justification of Selected Method

Median imputation is the preferred method for this dataset. The data contains extreme outliers (C4: Age = 105, Spend = 500) which heavily distort neighbour-based estimates — e.g. KNN pulls Monthly_Spend(C3) up to 400 purely because C4's inflated value of 500 sits in the same loyalty class. Median imputation is robust to such outliers, requires no distance computation, and is well suited to a small, noisy, manually-entered dataset. KNN is more appropriate for larger datasets with reliable feature correlations.

B. Outlier Detection and Treatment

(i) Z-score computation — Age (imputed dataset)

Values: 23, 36, 31, 105, 27, 40, 36, 42 → Mean = 42.5, Std. Dev. $\approx$ 24.37

Customer	Age	Z-score
C1	23	-0.80
C2	36	-0.27
C3	31	-0.47
C4	105	2.56
C5	27	-0.64
C6	40	-0.10
C7	36	-0.27
C8	42	-0.02

Z-score computation — Monthly_Spend (imputed dataset)

Values: 20, 35, 38, 500, 25, 45, 38, 300 → Mean = 125.13, Std. Dev. $\approx$ 166.55

Customer	Spend	Z-score
C1	20	-0.63
C2	35	-0.54
C3	38	-0.52
C4	500	2.25
C5	25	-0.60
C6	45	-0.48
C7	38	-0.52
C8	300	1.05

(ii) Extreme values identified

C4 — Age = 105 ($|Z| = 2.56$, exceeds the ± 2 threshold): biologically implausible age, almost certainly a data-entry error.

C4 — Monthly_Spend = ₹500K ($|Z| = 2.25$, exceeds ± 2): far above the rest of the distribution. Both extreme values occur in the same record (C4), suggesting a systematic entry error for that row rather than two independent outliers.

(iii) Treatment and Justification

Capping (Winsorization): cap Age(C4) to the next highest plausible value (e.g., 45, the dataset's second-highest realistic age) and cap Monthly_Spend(C4) to 300 (the next highest observed value), pulling extreme points to the edge of the acceptable range without discarding the record.

Log transformation: apply $\log(\text{Monthly_Spend})$ across the column to compress the scale and reduce the influence of the ₹500K value while preserving relative ordering.

Verification/removal: if source verification confirms Age = 105 is a typo (e.g., for 45) and Spend = 500 for 50), correcting at the source is preferable to capping; only remove the record if correction is impossible and the row would otherwise distort the recommendation model.

Capping is recommended here because it preserves sample size (important with only 8 records) while limiting the outlier's leverage on distance-based recommendation algorithms.

C. Categorical Data Standardization

(i) Inconsistencies identified in Gender

Raw values: M, Male, F, Female, male, FEMALE, M, Female — inconsistent casing and abbreviation (M vs Male, F vs Female, mixed case).

(ii) Standardized values

Customer	Original Gender	Standardized Gender
C1	M	Male
C2	Male	Male
C3	F	Female
C4	Female	Female
C5	male	Male
C6	FEMALE	Female
C7	M	Male
C8	Female	Female

Rule applied: case-normalize (e.g., .strip().title()) and map single-letter codes (M→Male, F→Female) to the two canonical categories Male / Female.

Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis — Income vs Purchase_Value

Mean(Income) = $645 / 8 = ₹80.625K$ Mean(Purchase_Value) = $1590 / 8 = ₹198.75K$

Cust	Income	PV	Income – mean	PV – mean	Product
U1	40	100	-40.63	-98.75	4011.72
U2	60	150	-20.63	-48.75	1005.47
U3	85	210	4.38	11.25	49.22
U4	120	320	39.38	121.25	4774.22
U5	45	110	-35.63	-88.75	3162.11
U6	75	180	-5.63	-18.75	105.47
U7	150	360	69.38	161.25	11185.55
U8	70	160	-10.63	-38.75	411.72

Sum of products = 24,705.48

Covariance (population) = $24,705.48 / 8 = 3,088.19$ (sample covariance, $/7 = 3,529.35$)

(ii) Interpretation

The covariance is large and positive, indicating that Income and Purchase_Value move together: customers with higher income tend to have higher purchase values.

(iii) Role of covariance in feature relationships

Covariance quantifies the direction (and, when scaled to correlation, the strength) of the linear relationship between two variables. A high positive covariance signals redundancy — the two features carry overlapping information — which is useful for deciding which attributes can be dropped or combined before modelling.

B. Feature Selection

(i) Redundant attributes

Income, Purchase Value, Avg Balance and EMI all rise together with a customer's overall financial capacity (visible in the monotonic increase from U1 → U7 in the table). Orders and Credit Score also trend with Customer Class but capture a behavioural /creditworthiness dimension rather than pure wealth. This overlap makes Income, Purchase Value, Avg Balance and EMI largely redundant with one another.

(ii) Optimal feature subset

Recommended subset: Income, Credit Score, Orders.

(iii) Justification

Income — representative summary of the customer's financial capacity (captures the signal otherwise duplicated in Purchase Value, Avg Balance, EMI).

Credit Score — an independent creditworthiness dimension not fully implied by income alone.

Orders — a behavioural engagement dimension distinct from pure wealth or credit standing.

This 3-feature subset reduces dimensionality and computational cost while preserving the distinct dimensions (wealth, credit risk, engagement) needed to classify Customer Class.

C. Data Transformation — Standardization (Z-score)

Column means and standard deviations:

Attribute	Mean	Std. Dev.
Age	39.375	10.22
Income	80.625	35.13
Purchase Value	198.75	88.52
Credit Score	723.125	54.25
Orders	25.25	8.04
Avg Balance	25.125	12.36
EMI	9500	4308.42

Standardized values ($z = (x - \text{mean}) / \text{std}$):

Cust	Age	Income	PV	Credit Score	Orders	Avg Bal	EMI
U1	-1.50	-1.16	-1.12	-1.35	-1.27	-1.06	-1.04
U2	-0.53	-0.59	-0.55	-0.43	-0.65	-0.58	-0.58
U3	0.45	0.12	0.13	-0.06	0.34	-0.01	-0.12
U4	1.24	1.12	1.37	1.42	1.21	1.20	1.28
U5	-1.02	-1.01	-1.00	-0.79	-0.90	-0.90	-0.93
U6	-0.04	-0.16	-0.21	-0.24	-0.16	-0.25	-0.23
U7	1.63	1.97	1.82	1.79	1.83	2.01	1.97
U8	-0.23	-0.30	-0.44	-0.33	-0.40	-0.41	-0.35

(ii) Why normalization / standardization is required

The raw attributes are on very different scales (EMI in thousands vs. Credit Score in hundreds vs. Orders in single/double digits). Distance-based and gradient-based models would let large-scale attributes (EMI, Purchase Value) dominate the objective purely due to magnitude, not actual importance. Standardizing to zero mean and unit variance puts all features on a comparable scale so each contributes fairly to the model.

Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning (3 bins)

Study Hours: range = $11 - 2 = 9$, bin width = $9 / 3 = 3 \rightarrow$ intervals [2,5), [5,8), [8,11]

Bin	Interval	Students (Study_Hours)
Bin 1	2 – 5	S1(2), S2(3), S3(4)
Bin 2	5 – 8	S4(5), S5(6), S6(7)
Bin 3	8 – 11	S7(8), S8(9), S9(10), S10(11)

Test Score: range = $100 - 40 = 60$, bin width = $60 / 3 = 20 \rightarrow$ intervals [40,60), [60,80), [80,100]

Bin	Interval	Students (Test Score)
Bin 1	40 – 60	S1(40), S2(50), S3(55)
Bin 2	60 – 80	S4(65), S5(75)
Bin 3	80 – 100	S6(85), S7(90), S8(95), S9(98), S10(100)

B. Equal-Frequency Binning (3 bins, ~equal records)

Bin	Study Hours	Test Score	Students
Bin 1	2, 3, 4	40, 50, 55	S1, S2, S3
Bin 2	5, 6, 7	65, 75, 85	S4, S5, S6
Bin 3	8, 9, 10, 11	90, 95, 98, 100	S7, S8, S9, S10

Each bin holds roughly the same number of records (3, 3, 4), unlike equal-width binning where record counts per bin can vary sharply.

C. Concept Hierarchy Construction

Study_Hours (hrs)	Concept Level	Test_Score	Concept Level
2 – 4	Low	40 – 59	Poor
5 – 7	Medium	60 – 79	Average
8 – 11	High	80 – 100	Excellent

This hierarchy maps directly onto the equal-width bins above and improves interpretability for the performance-prediction model (e.g., "High study hours + Excellent score → Excellent performance").

Q4. Normalization of {200, 300, 400, 600, 1000}

(a) Min-Max Normalization (min = 0, max = 1)

Formula: $X' = (X - X_{min}) / (X_{max} - X_{min})$, $X_{min} = 200$, $X_{max} = 1000$, range = 800

X	X'
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-score Normalization

Mean = $(200+300+400+600+1000)/5 = 500$

Variance = $[(-300)^2 + (-200)^2 + (-100)^2 + (100)^2 + (500)^2] / 5 = 400000 / 5 = 80,000 \rightarrow \text{Std.}$

Dev. ≈ 282.84

X	X – mean	Z = (X – mean)/std
200	-300	-1.06
300	-200	-0.71
400	-100	-0.35
600	100	0.35
1000	500	1.77

Q5. Data Transformation in Online Shopping Behaviour Analysis

A. Equal-Frequency Binning (3 bins)

Bin	Visit_Duration (s)	Customers
Bin 1 (short)	15, 30, 45	C1, C2, C3
Bin 2 (medium)	60, 120, 300	C4, C5, C6
Bin 3 (long)	600, 900, 1800, 3600	C7, C8, C9, C10

(iii) **Effectiveness:** equal-frequency binning keeps a balanced number of records per bin (3, 3, 4) even though the underlying values are heavily right-skewed. It successfully separates "quick glance" visits (Bin 1), "moderate browsing" visits (Bin 2) and "deep engagement" visits (Bin 3), which is useful for segmentation even though the value ranges within Bin 3 are still very wide (600–3600s).

B. Z-score Normalization

Mean = $7470 / 10 = 747$ seconds. Variance = $11,901,060 / 10 = 1,190,106 \rightarrow$ Std. Dev. ≈ 1090.92

Customer	Duration (s)	Z-score
C1	15	-0.67
C2	30	-0.66
C3	45	-0.64
C4	60	-0.63
C5	120	-0.57
C6	300	-0.41
C7	600	-0.13
C8	900	0.14
C9	1800	0.97
C10	3600	2.62

(iii) **Handling extreme values:** Z-score normalization does not bound the output to a fixed range, so extreme durations (1800s, 3600s) surface clearly as large positive z-scores (0.97 and

2.62). This preserves the information that these are genuine outliers relative to the group, but it also means the transformed scale is still technically unbounded — a clustering algorithm using Euclidean distance will still be influenced by these large z-values, just less severely than by the raw seconds.

C. Min-Max Normalization ([0, 1])

$X_{\min} = 15$, $X_{\max} = 3600$, range = 3585

Customer	Duration (s)	X' (Min-Max)
C1	15	0.000
C2	30	0.004
C3	45	0.008
C4	60	0.013
C5	120	0.029
C6	300	0.080
C7	600	0.163
C8	900	0.247
C9	1800	0.498
C10	3600	1.000

(iii) Comparison: Min-Max vs Z-score

Min-Max compresses almost all "normal" visit durations (15–900s) into the narrow band 0.00–0.25, because the single extreme value (3600s) stretches the whole scale — it is highly sensitive to outliers.

Z-score keeps outliers visible as large-magnitude values (e.g., 2.62 for 3600s) rather than compressing everything else, and is comparatively less distorted by a single extreme point, though it does not bound the range.

For clustering algorithms sensitive to scale, Z-score is generally preferred here, since Min-Max would make most customers look almost identical (all near 0) while a few dominate the scale near 1.